

Table 1: End-to-end execution time (s, median per iteration) and speedup split by pass type. When present, the OctTree row includes ColorOctree passes; a separate ColorOctree row reports only those passes. ADT fields = non-recursive fields annotated with `@BENCH adt_fields`; SoA bufs = 1+ non-recursive field slots across all constructors (recursive children are stored in the tag buffer). Speedup $>1\times$ means SoA is faster; **bold** marks $>1.1\times$.

Program	ADT	SoA	Fold passes			Map passes		
	fields	bufs	AoS (s)	SoA (s)	Speedup	AoS (s)	SoA (s)	Speedup
OctTree	16	9	0.4412	0.4911	0.90 $\times$	0.3084	0.3612	0.85 $\times$
Compiler	9	8	0.1602	0.0581	2.76$\times$	0.1687	0.1890	0.89 $\times$
DBQuery	15	13	0.1722	0.0939	1.83$\times$	0.1573	0.1930	0.81 $\times$
DecisionTree	7	6	0.5288	0.6363	0.83 $\times$	–	–	–
DomTree	15	14	0.1387	0.0619	2.24$\times$	0.0591	0.0816	0.72 $\times$
KDTree	17	16	0.2016	0.1449	1.39$\times$	–	–	–
LinearListReduction	11	11	0.0308	0.0057	5.40$\times$	–	–	–
List	2	2	0.2346	0.1118	2.10$\times$	0.2599	0.3402	0.76 $\times$
MonoTree	3	2	0.0178	0.0223	0.80 $\times$	0.0984	0.0796	1.24$\times$
ObjectGraph	5	4	0.1126	0.1273	0.88 $\times$	0.1427	0.1636	0.87 $\times$
PiecewiseFunctions	8	7	0.1435	0.0984	1.46$\times$	0.2393	0.2691	0.89 $\times$
TernaryTree	5	3	0.0287	0.0280	1.03 $\times$	0.1015	0.1135	0.89 $\times$
Trie	10	9	0.0658	0.1136	0.58 $\times$	0.1499	0.2349	0.64 $\times$
ColorOctree	25	18	0.0912	0.0796	1.14$\times$	–	–	–

Table 2: PAPI speedup summary across all passes. For each program and counter, speedup is AoS/SoA using summed per-pass median counters. $>1\times$ means SoA has fewer events.

Program	CYC	INS	L1D	L1I	L2D	L2I	LLC
OctTree	1.01 $\times$	0.75 $\times$	1.53$\times$	0.49 $\times$	1.97$\times$	0.61 $\times$	3.76$\times$
Compiler	1.76$\times$	0.52 $\times$	4.41$\times$	0.51 $\times$	14.52$\times$	0.25 $\times$	5.75$\times$
DBQuery	1.21$\times$	0.47 $\times$	0.86 $\times$	0.35 $\times$	6.47$\times$	0.20 $\times$	6.71$\times$
DecisionTree	0.83 $\times$	0.69 $\times$	0.79 $\times$	0.83 $\times$	0.44 $\times$	0.84 $\times$	0.93 $\times$
DomTree	1.49$\times$	0.57 $\times$	2.17$\times$	0.86 $\times$	21.06$\times$	0.28 $\times$	10.06$\times$
KDTree	1.37$\times$	0.77 $\times$	1.81$\times$	0.91 $\times$	4.60$\times$	0.86 $\times$	4.04$\times$
LinearListReduction	4.94$\times$	0.90 $\times$	1.92$\times$	2.96$\times$	6.37$\times$	4.04$\times$	10.41$\times$
List	0.74 $\times$	0.80 $\times$	2.29$\times$	0.57 $\times$	0.67 $\times$	1.70$\times$	1.28$\times$
MonoTree	0.98 $\times$	0.85 $\times$	1.01 $\times$	0.89 $\times$	0.68 $\times$	1.16$\times$	1.15$\times$
ObjectGraph	0.93 $\times$	0.75 $\times$	4.28$\times$	0.57 $\times$	0.86 $\times$	0.99 $\times$	2.74$\times$
PiecewiseFunctions	1.07 $\times$	0.60 $\times$	2.70$\times$	0.83 $\times$	3.97$\times$	0.58 $\times$	4.51$\times$
TernaryTree	1.13$\times$	0.79 $\times$	0.47 $\times$	0.55 $\times$	0.39 $\times$	0.94 $\times$	1.40$\times$
Trie	1.11$\times$	0.56 $\times$	1.97$\times$	0.66 $\times$	1.91$\times$	0.26 $\times$	7.20$\times$
Geomean	1.24$\times$	0.68 $\times$	1.68$\times$	0.72 $\times$	2.33$\times$	0.68 $\times$	3.47$\times$

Table 3: PAPI counter totals across all passes (AoS/SoA). Each entry is the sum of per-pass median counter values.

Program	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
OctTree	2.6e+09/ 2.6e+09	8.2e+09 /1.1e+10	3.6e+07/ 2.3e+07	4.0e+05 /8.2e+05	9.6e+05/ 4.9e+05	1.3e+04 /2.2e+04	6.3e+07/ 1.7e+07
Compiler	1.1e+09/ 6.0e+08	9.9e+08 /1.9e+09	1.3e+07/ 2.9e+06	2.6e+05 /5.1e+05	1.3e+06/ 8.7e+04	1.1e+04 /4.6e+04	5.1e+07/ 8.9e+06
DBQuery	1.2e+09/ 1.0e+09	1.1e+09 /2.2e+09	3.7e+06 /4.3e+06	1.6e+05 /4.5e+05	7.4e+05/ 1.1e+05	7.9e+03 /3.9e+04	2.3e+07/ 3.5e+06
DecisionTree	2.7e+09 /3.2e+09	6.3e+09 /9.0e+09	7.9e+07 /1.0e+08	1.1e+04 /1.3e+04	5.8e+05 /1.3e+06	1.1e+04 /1.3e+04	1.6e+08 /1.7e+08
DomTree	8.2e+08/ 5.5e+08	1.2e+09 /2.2e+09	1.2e+07/ 5.6e+06	9.7e+04 /1.1e+05	1.2e+06/ 5.8e+04	8.5e+03 /3.0e+04	5.7e+07/ 5.7e+06
KDTree	9.8e+08/ 7.2e+08	3.1e+09 /4.0e+09	1.9e+07/ 1.0e+07	1.7e+03 /1.9e+03	5.8e+05/ 1.3e+05	1.5e+03 /1.7e+03	4.3e+07/ 1.1e+07
LinearListReduction	1.4e+08/ 2.9e+07	9.0e+07 /1.0e+08	1.5e+06/ 7.9e+05	7.4e+02/ 2.5e+02	3.2e+05/ 5.0e+04	6.1e+02/ 1.5e+02	1.2e+07/ 1.2e+06
List	1.1e+09 /1.5e+09	4.5e+09 /5.6e+09	1.1e+07/ 4.9e+06	3.1e+05 /5.5e+05	1.3e+05 /2.0e+05	1.8e+04/ 1.0e+04	4.3e+07/ 3.3e+07
MonoTree	2.5e+08 /2.6e+08	9.1e+08 /1.1e+09	5.6e+05/ 5.5e+05	4.3e+04 /4.8e+04	1.0e+04 /1.5e+04	3.2e+03/ 2.7e+03	2.1e+06/ 1.8e+06
ObjectGraph	8.1e+08 /8.7e+08	3.0e+09 /4.0e+09	9.7e+06/ 2.3e+06	1.8e+05 /3.1e+05	6.2e+04 /7.2e+04	1.3e+04 /1.3e+04	2.2e+07/ 8.1e+06
PiecewiseFunctions	1.2e+09/ 1.2e+09	2.8e+09 /4.7e+09	1.7e+07/ 6.2e+06	3.0e+05 /3.6e+05	3.8e+05/ 9.6e+04	2.2e+04 /3.8e+04	4.4e+07/ 9.7e+06
TernaryTree	4.2e+08/ 3.7e+08	1.2e+09 /1.5e+09	6.3e+05 /1.3e+06	6.8e+04 /1.2e+05	8.4e+03 /2.1e+04	2.6e+03 /2.7e+03	3.2e+06/ 2.3e+06
Trie	6.4e+08/ 5.8e+08	1.2e+09 /2.2e+09	8.7e+06/ 4.4e+06	1.8e+05 /2.7e+05	1.6e+05/ 8.5e+04	7.8e+03 /3.0e+04	2.6e+07/ 3.6e+06

Table 4: Mutable vs immutable cursor comparison (times only). Times are median per iteration (s). **Bold** marks the fastest time across all four variants.

Program	Am	Ai	Sm	Si
Compiler	0.3289	1.206	0.2472	<i>OOM</i>
DBQuery	0.3294	0.3864	0.2869	1.136
DecisionTree	0.5288	0.9447	0.6363	3.245
DomTree	0.1978	0.3377	0.1435	0.5308
KDTree	0.2016	0.2528	0.1449	0.8482
LinearListReduction	0.0308	0.1330	0.0057	<i>OOM</i>
List	0.4945	<i>OOM</i>	0.4520	<i>OOM</i>
MonoTree	0.1161	0.1841	0.1019	0.2825
ObjectGraph	0.2553	0.5116	0.2910	0.7142
Oct:sumMass	0.0404	0.0491	0.0298	<i>OOM</i>
Oct:sumEnergy	0.0404	0.0661	0.0376	<i>OOM</i>
Oct:countActive	0.0444	0.0796	0.0311	<i>OOM</i>
Oct:countParticles	0.0539	0.0466	0.0557	<i>OOM</i>
Oct:barnesHutPotential	0.0431	0.0596	0.0443	<i>OOM</i>
Oct:fmmPotential	0.1279	0.0995	0.2130	<i>OOM</i>
Oct:scaleEnergy	0.1534	0.1536	0.1767	<i>OOM</i>
Oct:clearFlags	0.1550	0.1508	0.1845	<i>OOM</i>
PiecewiseFunctions	0.3828	0.5753	0.3675	2.199
TernaryTree	0.1302	0.1648	0.1414	0.6213
Trie	0.2157	0.3076	0.3485	1.249
ColorOctree	0.0912	0.1196	0.0796	<i>OOM</i>

Table 5: Mutable vs immutable cursor comparison (speedups). Shown: AoS-mut/SoA-mut, AoS-imm/AoS-mut, AoS-imm/SoA-mut, AoS-imm/SoA-imm. $>1\times$ means the denominator is faster.

Program	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
Compiler	1.33 $\times$	3.67 $\times$	4.88 $\times$	–
DBQuery	1.15 $\times$	1.17 $\times$	1.35 $\times$	0.34 $\times$
DecisionTree	0.83 $\times$	1.79 $\times$	1.48 $\times$	0.29 $\times$
DomTree	1.38 $\times$	1.71 $\times$	2.35 $\times$	0.64 $\times$
KDTree	1.39 $\times$	1.25 $\times$	1.75 $\times$	0.30 $\times$
LinearListReduction	5.40 $\times$	4.32 $\times$	23.34 $\times$	–
List	1.09 $\times$	–	–	–
MonoTree	1.14 $\times$	1.59 $\times$	1.81 $\times$	0.65 $\times$
ObjectGraph	0.88 $\times$	2.00 $\times$	1.76 $\times$	0.72 $\times$
Oct:sumMass	1.36 $\times$	1.22 $\times$	1.65 $\times$	–
Oct:sumEnergy	1.08 $\times$	1.63 $\times$	1.76 $\times$	–
Oct:countActive	1.43 $\times$	1.80 $\times$	2.56 $\times$	–
Oct:countParticles	0.97 $\times$	0.87 $\times$	0.84 $\times$	–
Oct:barnesHutPotential	0.97 $\times$	1.39 $\times$	1.35 $\times$	–
Oct:fmmPotential	0.60 $\times$	0.78 $\times$	0.47 $\times$	–
Oct:scaleEnergy	0.87 $\times$	1.00 $\times$	0.87 $\times$	–
Oct:clearFlags	0.84 $\times$	0.97 $\times$	0.82 $\times$	–
PiecewiseFunctions	1.04 $\times$	1.50 $\times$	1.57 $\times$	0.26 $\times$
TernaryTree	0.92 $\times$	1.27 $\times$	1.17 $\times$	0.27 $\times$
Trie	0.62 $\times$	1.43 $\times$	0.88 $\times$	0.25 $\times$
ColorOctree	1.14 $\times$	1.31 $\times$	1.50 $\times$	–

Table 6: Runtime comparison of Gibbon (AoS/SoA), GHC. Times are total median per iteration (s). **Bold** marks the fastest time for each program.

Program	Gibbon-AoS	Gibbon-SoA	GHC
Compiler	0.3289	0.2472	2.170
DBQuery	0.3294	0.2869	2.410
DecisionTree	0.5288	0.6363	4.113
DomTree	0.1978	0.1435	1.190
KDTree	0.2016	0.1449	1.582
LinearListReduction	0.0308	0.0057	1.264
List	0.4945	0.4520	16.99
MonoTree	0.1161	0.1019	1.506
ObjectGraph	0.2553	0.2910	2.508
Oct:sumMass	0.0404	0.0298	3.229
Oct:sumEnergy	0.0404	0.0376	6.008
Oct:countActive	0.0444	0.0311	5.245
Oct:countParticles	0.0539	0.0557	5.304
Oct:barnesHutPotential	0.0431	0.0443	5.346
Oct:fmmPotential	0.1279	0.2130	5.742
Oct:scaleEnergy	0.1534	0.1767	6.062
Oct:clearFlags	0.1550	0.1845	5.304
PiecewiseFunctions	0.3828	0.3675	3.017
TernaryTree	0.1302	0.1414	0.9950
Trie	0.2157	0.3485	1.695
ColorOctree	0.0912	0.0796	7.071
Geomean	0.1347	0.1217	3.268

Table 7: GHC speedups vs Gibbon mutable variants. Each entry is total median runtime speedup over all passes for one iteration. *GHC/AoS* and *GHC/SoA* are reported.

Program	GHC/AoS	GHC/SoA
Compiler	6.60×	8.78×
DBQuery	7.32×	8.40×
DecisionTree	7.78×	6.46×
DomTree	6.02×	8.29×
KDTree	7.84×	10.92×
LinearListReduction	41.07×	221.89×
List	34.35×	37.58×
MonoTree	12.96×	14.77×
ObjectGraph	9.82×	8.62×
Oct:sumMass	79.95×	108.37×
Oct:sumEnergy	148.55×	159.88×
Oct:countActive	118.22×	168.83×
Oct:countParticles	98.40×	95.14×
Oct:barnesHutPotential	124.15×	120.76×
Oct:fmmpotential	44.88×	26.95×
Oct:scaleEnergy	39.51×	34.32×
Oct:clearFlags	34.22×	28.75×
PiecewiseFunctions	7.88×	8.21×
TernaryTree	7.64×	7.04×
Trie	7.86×	4.86×
ColorOctree	77.57×	88.80×
Geomean	24.27×	26.85×

Table 8: Per-pass performance for `OctTree`, OctTree SoA uses 9 buffers; ColorOctree SoA uses 18 buffers (mutable + immutable cursors). Times are median per iteration (s); ± shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup >1× means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
barnesHutPotential	F	11/16	31%	0.0431	0.0596	0.0443	–	0.97×	1.39 ×	1.35 ×	–
clearFlags	M	15/16	6%	0.1550	0.1508	0.1845	–	0.84×	0.97×	0.82×	–
countActive	F	10/16	38%	0.0444	0.0796	0.0311	–	1.43 ×	1.80 ×	2.56 ×	–
countParticles	F	8/16	50%	0.0539	0.0466	0.0557	–	0.97×	0.87×	0.84×	–
fmmpotential	F	12/16	25%	0.1279	0.0995	0.2130	–	0.60×	0.78×	0.47×	–
scaleEnergy	M	16/16	0%	0.1534	0.1536	0.1767	–	0.87×	1.00×	0.87×	–
sumEnergy	F	12/16	25%	0.0404	0.0661	0.0376	–	1.08×	1.63 ×	1.76 ×	–
sumMass	F	10/16	38%	0.0404	0.0491	0.0298	–	1.36 ×	1.22 ×	1.65 ×	–
paletteEntriesQuantized	F	13/25	48%	0.0445	0.0560	0.0366	–	1.22 ×	1.26 ×	1.53 ×	–
quantizationErrorProxy	F	10/25	60%	0.0466	0.0637	0.0430	–	1.08×	1.37 ×	1.48 ×	–
Total				0.7497	0.8247	0.8523	–	0.88×	1.10 ×	0.97×	–
Geomean								1.01×	1.19 ×	1.20 ×	–

Table 9: Per-pass PAPI counters for `OctTree` (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
barnesHutPotential	F	2.1e+08 /2.2e+08	8.2e+08 /1.0e+09	4.2e+04 /5.7e+06	4.2e+02 /4.6e+02	1.8e+03 /1.1e+05	2.4e+02 /3.0e+02	6.2e+06/ 1.6e+06
clearFlags	M	3.7e+08 /4.4e+08	8.0e+08 /1.7e+09	1.5e+06 /2.1e+06	1.6e+05 /4.1e+05	2.6e+04 /3.4e+04	5.0e+03 /9.2e+03	5.5e+06/ 2.4e+06
countActive	F	2.1e+08/ 1.5e+08	5.8e+08/ 5.8e+08	7.3e+06/ 9.1e+05	5.3e+02/ 4.4e+02	2.9e+05/ 2.3e+04	3.9e+02/ 2.5e+02	6.0e+06/ 3.0e+05
countParticles	F	1.5e+08/ 1.4e+08	5.7e+08/ 5.7e+08	1.3e+06/ 3.1e+05	4.8e+02/ 4.1e+02	7.4e+03 /8.0e+03	3.1e+02/ 2.4e+02	5.3e+06/ 1.1e+05
fmmpotential	F	4.9e+08/ 4.8e+08	1.6e+09 /1.8e+09	2.3e+04 /8.3e+05	8.7e+02 /1.5e+03	1.8e+03 /2.9e+04	7.5e+02 /1.2e+03	4.0e+06/ 2.2e+06
scaleEnergy	M	3.7e+08 /4.4e+08	9.1e+08 /1.9e+09	3.9e+06/ 2.7e+06	2.4e+05 /4.0e+05	1.0e+05/ 4.1e+04	5.4e+03 /9.7e+03	5.4e+06/ 2.5e+06
sumEnergy	F	1.9e+08/ 1.9e+08	7.7e+08 /8.9e+08	4.0e+06 /5.7e+06	6.2e+02/ 4.5e+02	1.4e+05/ 1.1e+05	4.6e+02/ 2.6e+02	6.1e+06/ 2.3e+06
sumMass	F	2.0e+08/ 1.4e+08	5.7e+08 /6.3e+08	7.3e+06/ 1.0e+06	5.9e+02/ 3.7e+02	2.9e+05/ 2.2e+04	4.2e+02/ 2.6e+02	6.2e+06/ 1.2e+06
paletteEntriesQuantized	F	2.1e+08/ 1.8e+08	6.7e+08 /7.6e+08	6.0e+06/ 2.9e+06	5.9e+02/ 5.1e+02	7.7e+04/ 7.6e+04	3.8e+02/ 3.8e+02	9.1e+06/ 1.2e+06
quantizationErrorProxy	F	2.4e+08/ 2.2e+08	9.7e+08 /1.2e+09	4.4e+06/ 1.2e+06	1.2e+02 /4.0e+02	2.8e+04 /3.4e+04	1.0e+02 /3.9e+02	9.2e+06/ 3.0e+06
Total		2.6e+09/2.6e+09	8.2e+09/1.1e+10	3.6e+07/2.3e+07	4.0e+05/8.2e+05	9.6e+05/4.9e+05	1.3e+04/2.2e+04	6.3e+07/1.7e+07

Table 10: Per-pass performance for `Compiler`, ADT has 9 fields, SoA uses 8 buffers (mutable + immutable cursors). Times are median per iteration (s); ± shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup >1× means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
blockCountPass	F	2/9	78%	0.0142	0.0880	0.0021	<i>OOM</i>	6.89 ×	6.19 ×	42.65 ×	–
branchStatsPass	F	2/9	78%	0.0139	0.0878	0.0039	<i>OOM</i>	3.56 ×	6.30 ×	22.47 ×	–
castInstCountPass	F	3/9	67%	0.0201	0.0878	0.0014	<i>OOM</i>	14.28 ×	4.36 ×	62.32 ×	–
has cycle	F	4/9	56%	0.0372	0.1044	0.0338	<i>OOM</i>	1.10 ×	2.81 ×	3.09 ×	–
instCountPass	F	2/9	78%	0.0144	0.0882	0.0021	<i>OOM</i>	6.93 ×	6.14 ×	42.52 ×	–
latencyModelPass	F	3/9	67%	0.0142	0.0883	0.0033	<i>OOM</i>	4.31 ×	6.21 ×	26.77 ×	–
memoryOpStatsPass	F	3/9	67%	0.0141	0.0877	0.0041	<i>OOM</i>	3.41 ×	6.24 ×	21.25 ×	–
stripSideEffectsPass	M	7/9	22%	0.0842	0.1955	0.0931	<i>OOM</i>	0.90×	2.32 ×	2.10 ×	–
targetReturnPass	M	9/9	0%	0.0845	0.2138	0.0959	<i>OOM</i>	0.88×	2.53 ×	2.23 ×	–
throughputModelPass	F	3/9	67%	0.0142	0.0880	0.0034	<i>OOM</i>	4.20 ×	6.18 ×	25.96 ×	–
verifyIR	F	9/9	0%	0.0179	0.0768	0.0041	<i>OOM</i>	4.39 ×	4.30 ×	18.88 ×	–
Total				0.3289	1.206	0.2472	–	1.33 ×	3.67 ×	4.88 ×	–
Geomean								3.34 ×	4.56 ×	15.23 ×	–

Table 11: Per-pass PAPI counters for **Compiler** (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
blockCountPass	F	7.1e+07/ 1.0e+07	5.1e+07/ 4.5e+07	1.3e+06/ 6.0e+03	2.4e+02/ 6.8e+01	2.3e+05/ 1.1e+03	2.2e+02/ 5.8e+01	4.8e+06/ 4.7e+04
branchStatsPass	F	6.9e+07/ 1.9e+07	7.9e+07 /8.6e+07	8.2e+05/ 2.3e+04	2.2e+02/ 1.4e+02	4.9e+04/ 3.3e+03	1.9e+02/ 1.3e+02	4.7e+06/ 6.4e+05
castInstCountPass	F	1.0e+08/ 6.9e+06	5.1e+07/ 3.9e+07	1.3e+04 /3.2e+04	1.2e+02/ 4.5e+01	8.2e+02 /1.2e+03	9.5e+01/ 3.0e+01	4.7e+06/ 2.5e+04
has cycle	F	1.4e+08/ 9.6e+07	1.1e+08 /1.5e+08	2.4e+06/ 5.1e+05	2.8e+04 /8.6e+04	1.5e+05/ 2.3e+04	2.2e+03/ 5.0e+02	5.0e+06/ 1.6e+06
instCountPass	F	7.2e+07/ 1.0e+07	5.6e+07 /5.7e+07	1.3e+06/ 1.3e+04	2.4e+02/ 1.3e+02	2.2e+05/ 1.1e+03	2.1e+02/ 1.2e+02	4.8e+06/ 7.2e+04
latencyModelPass	F	7.1e+07/ 1.6e+07	6.2e+07 /6.9e+07	1.2e+06/ 3.5e+04	2.3e+02/ 7.2e+01	2.1e+05/ 3.8e+03	2.0e+02/ 5.7e+01	4.8e+06/ 6.9e+05
memoryOpStatsPass	F	7.0e+07/ 2.0e+07	8.5e+07 /9.1e+07	6.3e+05/ 3.3e+04	1.7e+02/ 1.0e+02	3.0e+04/ 3.2e+03	1.6e+02/ 9.4e+01	4.7e+06/ 6.4e+05
stripSideEffectsPass	M	1.5e+08 /1.9e+08	1.8e+08 /6.0e+08	1.7e+06/ 1.1e+06	1.3e+05 /1.9e+05	7.0e+03 /2.0e+04	5.3e+03 /2.1e+04	4.3e+06/ 1.8e+06
targetReturnPass	M	1.6e+08 /2.0e+08	1.7e+08 /6.3e+08	1.3e+06/ 1.1e+06	9.6e+04 /2.4e+05	5.9e+03 /2.1e+04	2.0e+03 /2.3e+04	3.7e+06/ 2.0e+06
throughputModelPass	F	7.1e+07/ 1.7e+07	6.2e+07 /6.9e+07	1.2e+06/ 3.2e+04	2.5e+02/ 1.2e+02	2.1e+05/ 3.6e+03	2.2e+02/ 8.1e+01	4.8e+06/ 7.0e+05
verifyIR	F	8.8e+07/ 2.0e+07	7.4e+07 /8.6e+07	9.8e+05/ 9.6e+04	8.4e+02/ 6.5e+02	1.4e+05/ 6.6e+03	6.5e+02/ 5.0e+02	4.7e+06/ 6.9e+05
Total		1.1e+09/6.0e+08	9.9e+08/1.9e+09	1.3e+07/2.9e+06	2.6e+05/5.1e+05	1.3e+06/8.7e+04	1.1e+04/4.6e+04	5.1e+07/8.9e+06

Table 12: Per-pass PAPI counters for **Compiler** (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
blockCountPass	F	4.4e+08/–	1.6e+08/–	2.3e+06/–	3.0e+02/–	1.0e+05/–	2.8e+02/–	1.4e+07/–
branchStatsPass	F	4.4e+08/–	1.8e+08/–	2.1e+06/–	2.9e+02/–	4.9e+04/–	2.7e+02/–	1.5e+07/–
castInstCountPass	F	4.4e+08/–	1.8e+08/–	2.1e+06/–	3.5e+02/–	5.0e+04/–	3.3e+02/–	1.5e+07/–
has cycle	F	5.2e+08/–	2.1e+08/–	2.5e+06/–	5.3e+02/–	7.4e+04/–	5.0e+02/–	1.7e+07/–
instCountPass	F	4.4e+08/–	1.6e+08/–	2.3e+06/–	4.7e+02/–	1.1e+05/–	4.5e+02/–	1.4e+07/–
latencyModelPass	F	4.4e+08/–	1.7e+08/–	2.1e+06/–	4.8e+02/–	5.0e+04/–	4.6e+02/–	1.4e+07/–
memoryOpStatsPass	F	4.4e+08/–	1.9e+08/–	2.1e+06/–	4.3e+02/–	4.7e+04/–	4.2e+02/–	1.5e+07/–
stripSideEffectsPass	M	7.0e+08/–	3.3e+08/–	1.8e+06/–	1.5e+05/–	1.7e+04/–	9.5e+03/–	2.3e+07/–
targetReturnPass	M	6.8e+08/–	3.2e+08/–	2.0e+06/–	2.2e+05/–	2.1e+04/–	1.5e+04/–	2.1e+07/–
throughputModelPass	F	4.4e+08/–	1.7e+08/–	2.1e+06/–	3.1e+02/–	5.1e+04/–	2.8e+02/–	1.5e+07/–
verifyIR	F	3.8e+08/–	1.2e+08/–	2.5e+06/–	5.4e+02/–	2.0e+05/–	4.8e+02/–	1.1e+07/–
Total		5.4e+09/–	2.2e+09/–	2.4e+07/–	3.8e+05/–	7.8e+05/–	2.8e+04/–	1.7e+08/–

Table 13: Per-pass performance for **DBQuery**, ADT has 15 fields, SoA uses 13 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
clearQueryFlags	M	14/15	7%	0.0729	0.0859	0.0829	0.2995	0.88 $\times$	1.18$\times$	1.04 $\times$	0.29 $\times$
countJoins	F	3/15	80%	0.0277	0.0345	0.0112	0.0855	2.47$\times$	1.24$\times$	3.07$\times$	0.40 $\times$
filterSelectivitySkew	F	5/15	67%	0.0281	0.0343	0.0126	0.0859	2.23$\times$	1.22$\times$	2.72$\times$	0.40 $\times$
hashJoinPressure	F	5/15	67%	0.0284	0.0382	0.0125	0.0859	2.27$\times$	1.35$\times$	3.06$\times$	0.45 $\times$
scaleCosts	M	15/15	0%	0.0844	0.0891	0.1100	0.2994	0.77 $\times$	1.06 $\times$	0.81 $\times$	0.30 $\times$
sumCost	F	6/15	60%	0.0260	0.0342	0.0209	0.1055	1.24$\times$	1.32$\times$	1.64$\times$	0.32 $\times$
sumMemory	F	6/15	60%	0.0275	0.0344	0.0142	0.0875	1.94$\times$	1.25$\times$	2.42$\times$	0.39 $\times$
sumRows	F	6/15	60%	0.0345	0.0360	0.0225	0.0871	1.53$\times$	1.04 $\times$	1.60$\times$	0.41 $\times$
Total				0.3294	0.3864	0.2869	1.136	1.15$\times$	1.17$\times$	1.35$\times$	0.34 $\times$
Geomean								1.54$\times$	1.20$\times$	1.85$\times$	0.37 $\times$

Table 14: Per-pass PAPI counters for **DBQuery** (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
clearQueryFlags	M	1.7e+08 /2.2e+08	1.9e+08 /7.2e+08	2.4e+05 /1.4e+06	7.8e+04 /2.5e+05	2.8e+04/ 2.5e+04	2.7e+03 /1.4e+04	2.4e+06/ 9.3e+05
countJoins	F	1.4e+08/ 5.7e+07	7.4e+07/ 7.1e+07	6.1e+05/ 2.6e+04	3.1e+02/ 2.0e+02	1.5e+05/ 2.0e+03	2.4e+02/ 1.7e+02	3.1e+06/ 5.8e+04
filterSelectivitySkew	F	1.4e+08/ 6.3e+07	1.0e+08 /1.1e+08	4.4e+05/ 7.2e+04	2.8e+02/ 1.5e+02	1.0e+05/ 6.3e+03	2.6e+02/ 1.4e+02	3.1e+06/ 3.2e+05
hashJoinPressure	F	1.4e+08/ 6.3e+07	8.4e+07 /9.9e+07	6.1e+05/ 9.2e+04	3.3e+02/ 1.3e+02	1.6e+05/ 7.0e+03	3.0e+02/ 1.1e+02	3.1e+06/ 1.4e+05
scaleCosts	M	1.9e+08 /3.5e+08	2.2e+08 /7.5e+08	2.4e+05 /1.2e+06	7.7e+04 /2.0e+05	3.2e+04/ 2.7e+04	3.3e+03 /2.3e+04	2.8e+06/ 1.1e+06
sumCost	F	1.3e+08/ 7.3e+07	9.0e+07 /1.2e+08	8.7e+05 /9.9e+05	5.7e+02/ 4.5e+02	1.5e+05/ 2.3e+04	4.3e+02/ 3.4e+02	3.1e+06/ 3.0e+05
sumMemory	F	1.4e+08/ 7.2e+07	9.0e+07 /1.2e+08	4.9e+05/ 8.6e+04	4.0e+02/ 1.6e+02	1.1e+05/ 1.1e+04	3.7e+02/ 1.5e+02	3.1e+06/ 3.9e+05
sumRows	F	1.7e+08/ 1.1e+08	2.1e+08 /2.3e+08	2.3e+05 /4.4e+05	3.0e+02/ 1.8e+02	2.2e+04/ 1.3e+04	2.8e+02/ 1.7e+02	2.8e+06/ 3.0e+05
Total		1.2e+09/1.0e+09	1.1e+09/2.2e+09	3.7e+06/4.3e+06	1.6e+05/4.5e+05	7.4e+05/1.1e+05	7.9e+03/3.9e+04	2.3e+07/3.5e+06

Table 15: Per-pass PAPI counters for **DBQuery** (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
clearQueryFlags	M	2.1e+08 /1.3e+09	3.2e+08 /2.4e+09	5.6e+05 /1.9e+07	6.7e+04 /5.0e+05	1.5e+04 /6.0e+04	2.1e+03 /4.8e+04	2.3e+06/ 9.4e+05
countJoins	F	1.7e+08 /4.3e+08	1.6e+08 /7.7e+08	1.8e+06/ 3.3e+05	3.0e+02 /9.0e+02	1.0e+05/ 1.3e+04	2.7e+02/ 1.8e+02	2.7e+06/ 8.3e+04
filterSelectivitySkew	F	1.7e+08 /4.4e+08	2.0e+08 /7.9e+08	1.4e+06/ 8.8e+05	2.4e+02 /8.2e+02	7.7e+04/ 1.7e+04	2.1e+02 /2.9e+02	2.7e+06/ 2.2e+05
hashJoinPressure	F	1.9e+08 /4.4e+08	1.8e+08 /7.7e+08	1.5e+06/ 6.7e+05	2.8e+02 /6.7e+02	9.3e+04/ 1.6e+04	2.5e+02 /2.5e+02	2.6e+06/ 1.6e+05
scaleCosts	M	2.2e+08 /1.3e+09	3.5e+08 /2.4e+09	4.9e+05 /2.4e+07	1.1e+05 /5.2e+05	2.0e+04 /6.5e+04	3.4e+03 /5.0e+04	2.6e+06/ 1.2e+06
sumCost	F	1.7e+08 /4.4e+08	1.8e+08 /7.8e+08	1.4e+06/ 1.2e+06	5.1e+02 /1.5e+03	7.1e+04/ 2.1e+04	3.8e+02 /5.8e+02	2.6e+06/ 3.5e+05
sumMemory	F	1.7e+08 /4.4e+08	1.8e+08 /7.8e+08	1.5e+06/ 1.2e+06	3.2e+02 /8.6e+02	7.6e+04/ 2.0e+04	3.2e+02 /5.5e+02	2.6e+06/ 3.3e+05
sumRows	F	1.8e+08 /4.4e+08	2.3e+08 /8.3e+08	1.5e+06/ 1.2e+06	3.3e+02 /9.9e+02	6.8e+04/ 2.1e+04	3.0e+02 /5.2e+02	2.6e+06/ 3.5e+05
Total		1.5e+09/5.2e+09	1.8e+09/9.5e+09	1.0e+07/4.8e+07	1.8e+05/1.0e+06	5.2e+05/2.3e+05	7.3e+03/1.0e+05	2.1e+07/3.6e+06

Table 16: Per-pass performance for `DecisionTree`, ADT has 7 fields, SoA uses 6 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
classify Batch	F	5/7	29%	1.80e-05	5.30e-05	6.20e-05	1.59e-04	0.29 $\times$	2.94$\times$	0.85 $\times$	0.33 $\times$
classify Depth	F	4/7	43%	–	–	–	1.00e-06	–	–	–	–
classify tree	F	5/7	29%	–	2.00e-06	2.00e-06	5.00e-06	–	–	1.00 $\times$	0.40 $\times$
countFeatureUses	F	3/7	57%	0.0598	0.1033	0.0739	0.4059	0.81 $\times$	1.73$\times$	1.40$\times$	0.25 $\times$
countLeaves	F	2/7	71%	0.0573	0.1149	0.0665	0.3078	0.86 $\times$	2.00$\times$	1.73$\times$	0.37 $\times$
countNodes	F	2/7	71%	0.0589	0.1145	0.0681	0.3094	0.87 $\times$	1.94$\times$	1.68$\times$	0.37 $\times$
countSmallLeaves	F	3/7	57%	0.0600	0.1000	0.0718	0.3140	0.84 $\times$	1.67$\times$	1.39$\times$	0.32 $\times$
inferenceCost	F	2/7	71%	0.0577	0.1008	0.0672	0.3124	0.86 $\times$	1.75$\times$	1.50$\times$	0.32 $\times$
sumImpurity	F	3/7	57%	0.0590	0.1039	0.0718	0.4424	0.82 $\times$	1.76$\times$	1.45$\times$	0.23 $\times$
sumPathLengths	F	3/7	57%	0.0592	0.1038	0.0763	0.4135	0.78 $\times$	1.75$\times$	1.36$\times$	0.25 $\times$
sumSamples	F	3/7	57%	0.0585	0.1003	0.0737	0.4263	0.79 $\times$	1.71$\times$	1.36$\times$	0.24 $\times$
treeDepth	F	2/7	71%	0.0584	0.1033	0.0671	0.3135	0.87 $\times$	1.77$\times$	1.54$\times$	0.33 $\times$
Total				0.5288	0.9447	0.6363	3.245	0.83 $\times$	1.79$\times$	1.48$\times$	0.29 $\times$
Geomean								0.75 $\times$	1.88$\times$	1.36$\times$	0.31 $\times$

Table 17: Per-pass PAPI counters for `DecisionTree` (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
classify Batch	F	5.0e+04 /2.8e+05	1.7e+05 /2.7e+05	5.8e+01 /1.6e+03	2.7e+02 /3.3e+02	7.0e+00 /1.2e+03	6.8e+01 /6.9e+01	7.7e+01 /6.1e+03
classify Depth	F	8.5e+02 /2.4e+03	1.3e+03 /2.3e+03	9.0e+00 /2.8e+01	3.3e+01 /4.2e+01	0.0e+00 /2.0e+00	8.0e+00 /2.5e+01	7.0e+00 /6.2e+01
classify tree	F	1.4e+03 /1.2e+04	1.3e+03 /2.2e+03	2.0e+01 /1.1e+02	2.7e+01 /3.2e+01	9.0e+00 /8.9e+01	8.0e+00/ 6.0e+00	3.4e+01 /3.5e+02
countFeatureUses	F	3.0e+08 /3.7e+08	7.4e+08 /1.1e+09	9.0e+06 /1.1e+07	9.8e+02 /1.3e+03	4.7e+04 /1.4e+05	9.5e+02 /1.3e+03	1.8e+07 /2.0e+07
countLeaves	F	2.9e+08 /3.4e+08	6.5e+08 /9.3e+08	8.6e+06 /1.0e+07	1.2e+03/ 1.2e+03	8.8e+04 /1.4e+05	1.2e+03/ 1.2e+03	1.8e+07 /1.9e+07
countNodes	F	2.9e+08 /3.4e+08	6.5e+08 /9.3e+08	8.5e+06 /1.0e+07	1.8e+03/ 1.7e+03	1.0e+05 /2.0e+05	1.7e+03/ 1.6e+03	1.8e+07 /1.9e+07
countSmallLeaves	F	3.0e+08 /3.6e+08	7.1e+08 /1.0e+09	8.9e+06 /1.2e+07	1.0e+03 /1.5e+03	3.4e+04 /1.5e+05	9.8e+02 /1.4e+03	1.8e+07 /2.0e+07
inferenceCost	F	2.9e+08 /3.4e+08	6.9e+08 /9.8e+08	8.7e+06 /1.0e+07	1.1e+03 /1.8e+03	8.2e+04 /1.4e+05	1.1e+03 /1.8e+03	1.8e+07 /1.9e+07
sumImpurity	F	3.0e+08 /3.6e+08	6.8e+08 /1.0e+09	8.6e+06 /1.1e+07	8.6e+02 /1.3e+03	5.6e+04 /1.3e+05	8.3e+02 /1.3e+03	1.8e+07 /2.0e+07
sumPathLengths	F	3.0e+08 /3.9e+08	7.8e+08 /1.1e+09	9.4e+06 /1.3e+07	1.6e+03 /1.6e+03	5.0e+04 /1.4e+05	1.6e+03 /1.6e+03	1.8e+07 /2.0e+07
sumSamples	F	3.0e+08 /3.7e+08	6.6e+08 /9.7e+08	8.5e+06 /1.2e+07	1.1e+03 /1.3e+03	5.1e+04 /1.5e+05	1.1e+03 /1.2e+03	1.8e+07 /2.0e+07
treeDepth	F	3.0e+08 /3.4e+08	6.9e+08 /9.8e+08	8.9e+06 /1.0e+07	1.1e+03 /1.3e+03	6.9e+04 /1.4e+05	1.1e+03 /1.3e+03	1.8e+07 /1.9e+07
Total		2.7e+09/3.2e+09	6.3e+09/9.0e+09	7.9e+07/1.0e+08	1.1e+04/1.3e+04	5.8e+05/1.3e+06	1.1e+04/1.3e+04	1.6e+08/1.7e+08

Table 18: Per-pass PAPI counters for `DecisionTree` (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
classify Batch	F	1.9e+05/ 1.8e+05	2.0e+05 /2.9e+05	6.3e+02 /2.3e+03	3.2e+02 /3.5e+02	4.2e+02 /1.2e+03	6.8e+01/ 5.4e+01	2.0e+03 /5.5e+03
classify Depth	F	2.1e+03 /2.2e+03	1.5e+03 /2.3e+03	2.2e+01 /4.2e+01	4.8e+01 /5.3e+01	0.0e+00/0.0e+00	1.5e+01 /3.1e+01	1.9e+01 /5.5e+01
classify tree	F	9.3e+03/ 7.5e+03	1.5e+03 /2.3e+03	5.8e+01 /1.6e+02	3.2e+01 /3.9e+01	3.7e+01 /8.7e+01	1.2e+01/ 7.0e+00	1.2e+02 /4.0e+02
countFeatureUses	F	5.2e+08 /1.6e+09	1.4e+09 /2.9e+09	4.6e+05 /3.9e+06	5.9e+02 /5.3e+03	5.2e+03 /2.3e+05	5.8e+02 /5.0e+03	1.5e+07/ 1.1e+07
countLeaves	F	5.7e+08 /1.6e+09	1.3e+09 /2.8e+09	1.7e+07/ 1.5e+06	5.6e+02 /2.0e+03	4.8e+05/ 1.8e+05	5.3e+02 /2.0e+03	1.3e+07/ 1.0e+07
countNodes	F	5.7e+08 /1.6e+09	1.3e+09 /2.9e+09	1.7e+07/ 2.0e+06	1.1e+03 /2.5e+03	4.9e+05/ 1.9e+05	9.4e+02 /2.4e+03	1.3e+07/ 1.0e+07
countSmallLeaves	F	5.0e+08 /1.4e+09	1.3e+09 /2.8e+09	3.7e+05 /2.5e+06	5.0e+02 /1.8e+03	5.8e+03 /1.9e+05	4.7e+02 /1.8e+03	1.5e+07/ 1.2e+07
inferenceCost	F	5.1e+08 /1.6e+09	1.4e+09 /2.9e+09	5.4e+05 /1.7e+06	3.2e+02 /2.7e+03	6.2e+03 /1.9e+05	3.1e+02 /2.7e+03	1.5e+07/ 1.0e+07
sumImpurity	F	5.2e+08 /1.6e+09	1.4e+09 /2.9e+09	4.9e+05 /3.8e+06	6.2e+02 /4.3e+03	5.4e+03 /2.1e+05	5.8e+02 /4.0e+03	1.5e+07/ 1.0e+07
sumPathLengths	F	5.3e+08 /1.5e+09	1.4e+09 /3.0e+09	2.9e+05 /4.5e+06	3.2e+02 /2.0e+03	7.4e+03 /2.3e+05	3.1e+02 /1.9e+03	1.4e+07/ 1.1e+07
sumSamples	F	5.0e+08 /1.5e+09	1.3e+09 /2.9e+09	4.1e+05 /7.1e+06	3.6e+02 /7.4e+03	5.3e+03 /3.0e+05	3.3e+02 /6.9e+03	1.5e+07/ 1.1e+07
treeDepth	F	5.2e+08 /1.6e+09	1.4e+09 /2.9e+09	2.4e+06/ 1.7e+06	8.6e+02 /2.7e+03	6.1e+04 /1.9e+05	8.4e+02 /2.6e+03	1.4e+07/ 1.0e+07
Total		4.7e+09/1.4e+10	1.2e+10/2.6e+10	3.9e+07/2.9e+07	5.6e+03/3.1e+04	1.1e+06/1.9e+06	5.0e+03/2.9e+04	1.3e+08/9.6e+07

Table 19: Per-pass performance for `DomTree`, ADT has 15 fields, SoA uses 14 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
SumArea	F	6/15	60%	0.0366	0.0707	0.0201	0.0913	1.82$\times$	1.93$\times$	3.51$\times$	0.77 $\times$
computeWidths	M	14/15	7%	0.0298	0.0303	0.0448	0.0888	0.66 $\times$	1.02 $\times$	0.68 $\times$	0.34 $\times$
count styled	F	3/15	80%	0.0340	0.0687	0.0117	0.0879	2.91$\times$	2.02$\times$	5.88$\times$	0.78 $\times$
find max Bottom	F	5/15	67%	0.0342	0.0683	0.0184	0.0875	1.86$\times$	2.00$\times$	3.71$\times$	0.78 $\times$
scaleLayout	M	15/15	0%	0.0294	0.0304	0.0368	0.0880	0.80 $\times$	1.04 $\times$	0.83 $\times$	0.35 $\times$
sumTextWidth	F	3/15	80%	0.0338	0.0693	0.0116	0.0873	2.91$\times$	2.05$\times$	5.95$\times$	0.79 $\times$
Total				0.1978	0.3377	0.1435	0.5308	1.38$\times$	1.71$\times$	2.35$\times$	0.64 $\times$
Geomean								1.57$\times$	1.60$\times$	2.52$\times$	0.59 $\times$

Table 20: Per-pass PAPI counters for `DomTree` (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
SumArea	F	1.8e+08/ 9.8e+07	2.9e+08 /4.9e+08	3.0e+06/ 1.1e+06	1.2e+03/ 3.9e+02	3.2e+05/ 2.2e+04	1.1e+03/ 2.4e+02	1.4e+07/ 1.8e+06
computeWidths	M	6.5e+07 /1.4e+08	1.4e+08 /4.6e+08	1.2e+06 /1.6e+06	5.4e+04/ 4.7e+04	5.2e+03 /9.7e+03	2.0e+03 /1.5e+04	1.3e+06/ 4.5e+05
count styled	F	1.7e+08/ 5.9e+07	2.6e+08 /2.6e+08	2.1e+06/ 3.4e+04	7.9e+02/ 1.6e+02	3.0e+05/ 4.9e+03	7.5e+02/ 1.4e+02	1.4e+07/ 7.1e+05
find max Bottom	F	1.7e+08/ 9.3e+07	2.5e+08 /4.1e+08	2.5e+06/ 5.9e+05	7.8e+02/ 1.6e+02	1.5e+05/ 9.3e+03	7.3e+02/ 1.2e+02	1.4e+07/ 1.6e+06
scaleLayout	M	6.0e+07 /1.0e+08	8.8e+07 /2.9e+08	1.1e+06 /2.3e+06	4.0e+04 /6.4e+04	2.5e+04/ 7.8e+03	3.0e+03 /1.4e+04	1.4e+06/ 4.5e+05
sumTextWidth	F	1.7e+08/ 5.9e+07	2.0e+08 /2.4e+08	2.4e+06/ 2.0e+04	9.8e+02/ 2.0e+02	4.2e+05/ 4.3e+03	9.4e+02/ 1.8e+02	1.4e+07/ 7.2e+05
Total		8.2e+08/5.5e+08	1.2e+09/2.2e+09	1.2e+07/5.6e+06	9.7e+04/1.1e+05	1.2e+06/5.8e+04	8.5e+03/3.0e+04	5.7e+07/5.7e+06

Table 21: Per-pass PAPI counters for `DomTree` (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
SumArea	F	3.5e+08 /4.6e+08	4.4e+08 /1.8e+09	8.6e+06/ 3.0e+06	1.6e+03/ 1.3e+03	4.3e+04 /5.3e+04	1.5e+03/ 1.1e+03	1.0e+07/ 1.3e+06
computeWidths	M	6.8e+07 /3.6e+08	1.5e+08 /8.9e+08	1.3e+06 /3.1e+06	4.8e+04 /3.0e+05	3.8e+03 /1.4e+04	8.9e+02 /2.3e+04	1.2e+06/ 4.7e+05
count styled	F	3.5e+08 /4.5e+08	4.3e+08 /1.7e+09	1.0e+07/ 1.3e+06	1.8e+02 /6.6e+02	3.8e+04/ 3.7e+04	1.6e+02 /4.9e+02	1.0e+07/ 4.6e+05
find max Bottom	F	3.5e+08 /4.4e+08	4.6e+08 /1.8e+09	1.0e+07/ 1.5e+06	1.9e+02 /7.5e+02	1.7e+04 /2.8e+04	1.6e+02 /7.4e+02	1.0e+07/ 1.0e+06
scaleLayout	M	6.8e+07 /3.6e+08	1.1e+08 /7.6e+08	9.5e+05 /4.0e+06	3.6e+04 /1.4e+05	9.3e+03 /1.4e+04	8.9e+02 /2.2e+04	1.2e+06/ 4.6e+05
sumTextWidth	F	3.5e+08 /4.4e+08	3.9e+08 /1.7e+09	1.1e+07/ 1.3e+06	2.5e+02 /9.7e+02	8.7e+04/ 3.9e+04	2.3e+02 /8.0e+02	1.0e+07/ 4.5e+05
Total		1.5e+09/2.5e+09	2.0e+09/8.6e+09	4.2e+07/1.4e+07	8.6e+04/4.5e+05	2.0e+05/1.8e+05	3.8e+03/4.8e+04	4.3e+07/4.2e+06

Table 22: Per-pass performance for **KDTree**, ADT has 17 fields, SoA uses 16 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across $-iterate$ runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
Find nearest Neighbour	F	13/17	24%	0.0370	0.0467	0.0248	0.2498	1.49 $\times$	1.26 $\times$	1.89 $\times$	0.19 $\times$
countInRange tight_box	F	13/17	24%	0.0311	0.0401	0.0202	0.1217	1.53 $\times$	1.29 $\times$	1.98 $\times$	0.33 $\times$
photonMappingBenchmark	F	12/17	29%	0.0452	0.0476	0.0414	0.1302	1.09 $\times$	1.05 $\times$	1.15 $\times$	0.37 $\times$
pointCloudNeighborhood	F	11/17	35%	0.0273	0.0389	0.0171	0.1132	1.60 $\times$	1.42 $\times$	2.28 $\times$	0.34 $\times$
sumMassInRange	F	14/17	18%	0.0296	0.0403	0.0205	0.1213	1.45 $\times$	1.36 $\times$	1.97 $\times$	0.33 $\times$
twoPointCorrelation bin_8_16	F	11/17	35%	0.0315	0.0391	0.0209	0.1120	1.51 $\times$	1.24 $\times$	1.87 $\times$	0.35 $\times$
Total				0.2016	0.2528	0.1449	0.8482	1.39 $\times$	1.25 $\times$	1.75 $\times$	0.30 $\times$
Geomean								1.43 $\times$	1.27 $\times$	1.82 $\times$	0.31 $\times$

Table 23: Per-pass PAPI counters for **KDTree** (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
Find nearest Neighbour	F	1.5e+08/ 1.1e+08	4.7e+08 /6.3e+08	6.4e+06/ 2.4e+06	4.5e+02 /4.6e+02	4.7e+05/ 3.5e+04	3.2e+02 /3.4e+02	6.9e+06/ 1.8e+06
countInRange tight_box	F	1.6e+08/ 1.0e+08	4.0e+08 /5.5e+08	2.8e+06/ 1.7e+06	2.4e+02 /2.7e+02	9.1e+04/ 2.0e+04	2.2e+02 /2.7e+02	7.5e+06/ 1.9e+06
photonMappingBenchmark	F	2.3e+08/ 2.1e+08	1.0e+09 /1.2e+09	2.2e+06 /2.5e+06	4.4e+02 /5.6e+02	8.6e+03 /1.9e+04	4.2e+02 /5.4e+02	5.6e+06/ 1.5e+06
pointCloudNeighborhood	F	1.4e+08/ 8.5e+07	3.3e+08 /4.8e+08	2.7e+06/ 9.8e+05	2.4e+02/ 2.0e+02	5.2e+03 /1.5e+04	2.2e+02/ 1.9e+02	7.7e+06/ 1.7e+06
sumMassInRange	F	1.5e+08/ 1.0e+08	4.0e+08 /5.7e+08	2.5e+06/ 2.1e+06	1.4e+02 /2.6e+02	3.9e+03 /2.4e+04	1.1e+02 /2.5e+02	7.6e+06/ 2.3e+06
twoPointCorrelation bin_8_16	F	1.6e+08/ 1.1e+08	4.8e+08 /6.1e+08	2.1e+06/ 6.0e+05	2.0e+02/ 1.4e+02	2.8e+03 /1.3e+04	1.8e+02/ 1.2e+02	7.6e+06/ 1.4e+06
Total		9.8e+08/7.2e+08	3.1e+09/4.0e+09	1.9e+07/1.0e+07	1.7e+03/1.9e+03	5.8e+05/1.3e+05	1.5e+03/1.7e+03	4.3e+07/1.1e+07

Table 24: Per-pass PAPI counters for **KDTree** (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
Find nearest Neighbour	F	2.1e+08 /5.9e+08	4.9e+08 /1.3e+09	5.7e+06/ 5.7e+06	5.9e+02 /3.7e+03	1.1e+05/ 4.7e+04	4.3e+02 /2.5e+03	5.5e+06/ 1.5e+06
countInRange tight_box	F	2.0e+08 /6.2e+08	4.5e+08 /1.3e+09	5.1e+06 /5.8e+06	2.6e+02 /1.8e+03	6.3e+03 /2.3e+04	2.4e+02 /1.8e+03	5.8e+06/ 1.6e+06
photonMappingBenchmark	F	2.4e+08 /6.6e+08	1.1e+09 /2.0e+09	3.0e+06 /5.8e+06	3.2e+02 /3.0e+03	3.2e+03 /2.4e+04	3.0e+02 /2.9e+03	5.2e+06/ 1.5e+06
pointCloudNeighborhood	F	2.0e+08 /5.7e+08	4.1e+08 /1.3e+09	5.1e+06/ 4.7e+06	2.7e+02 /1.5e+03	1.7e+04 /1.9e+04	2.5e+02 /1.4e+03	5.9e+06/ 1.4e+06
sumMassInRange	F	2.0e+08 /6.1e+08	4.7e+08 /1.3e+09	5.0e+06 /6.3e+06	2.0e+02 /1.9e+03	5.9e+03 /2.5e+04	1.9e+02 /1.9e+03	5.8e+06/ 1.6e+06
twoPointCorrelation bin_8_16	F	2.0e+08 /5.6e+08	6.1e+08 /1.4e+09	4.7e+06/ 3.6e+06	2.9e+02 /1.2e+03	6.1e+03 /1.8e+04	2.8e+02 /1.2e+03	5.9e+06/ 1.3e+06
Total		1.3e+09/3.6e+09	3.5e+09/8.4e+09	2.9e+07/3.2e+07	1.9e+03/1.3e+04	1.4e+05/1.6e+05	1.7e+03/1.2e+04	3.4e+07/8.9e+06

Table 25: Per-pass performance for **LinearListReduction**, ADT has 11 fields, SoA uses 11 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across $-iterate$ runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
reduction	F	2/11	82%	0.0308	0.1330	0.0057	<i>OOM</i>	5.40 $\times$	4.32 $\times$	23.34 $\times$	–
Total				0.0308	0.1330	0.0057	–	5.40 $\times$	4.32 $\times$	23.34 $\times$	–
Geomean								5.40 $\times$	4.32 $\times$	23.34 $\times$	–

Table 26: Per-pass PAPI counters for **LinearListReduction** (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
reduction	F	1.4e+08/ 2.9e+07	9.0e+07 /1.0e+08	1.5e+06/ 7.9e+05	7.4e+02/ 2.5e+02	3.2e+05/ 5.0e+04	6.1e+02/ 1.5e+02	1.2e+07/ 1.2e+06
Total		1.4e+08/2.9e+07	9.0e+07/1.0e+08	1.5e+06/7.9e+05	7.4e+02/2.5e+02	3.2e+05/5.0e+04	6.1e+02/1.5e+02	1.2e+07/1.2e+06

Table 27: Per-pass PAPI counters for **LinearListReduction** (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
reduction	F	6.7e+08/–	2.5e+08/–	4.0e+06/–	6.5e+02/–	4.2e+05/–	5.1e+02/–	2.7e+07/–
Total		6.7e+08/–	2.5e+08/–	4.0e+06/–	6.5e+02/–	4.2e+05/–	5.1e+02/–	2.7e+07/–

Table 28: Per-pass performance for **List**, ADT has 2 fields, SoA uses 2 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across $-iterate$ runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
add1 List	M	2/2	0%	0.2599	<i>OOM</i>	0.3402	<i>OOM</i>	0.76 $\times$	–	–	–
length List	F	1/2	50%	0.1315	<i>OOM</i>	0.0218	<i>OOM</i>	6.04 $\times$	–	–	–
sumList	F	2/2	0%	0.0585	<i>OOM</i>	0.0450	<i>OOM</i>	1.30 $\times$	–	–	–
sumList tail recursive	F	2/2	0%	0.0447	<i>OOM</i>	0.0451	<i>OOM</i>	0.99 $\times$	–	–	–
Total				0.4945	–	0.4520	–	1.09 $\times$	–	–	–
Geomean								1.56 $\times$	–	–	–

Table 29: Per-pass PAPI counters for **List** (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
add1 List	M	5.9e+08 /9.8e+08	1.9e+09 /2.8e+09	8.1e+06/ 3.7e+06	3.1e+05 /5.5e+05	2.7e+04 /5.9e+04	1.7e+04/ 1.0e+04	9.6e+06/ 7.8e+06
length List	F	1.2e+08/ 1.1e+08	8.0e+08/ 8.0e+08	3.5e+05/ 1.1e+05	1.8e+02/ 1.4e+02	4.3e+03 /1.4e+04	1.7e+02/ 1.2e+02	8.4e+06/ 1.2e+06
sumList	F	2.0e+08 /2.3e+08	9.0e+08 /1.0e+09	2.3e+06/ 5.4e+05	3.9e+02/ 1.3e+02	9.9e+04/ 6.9e+04	3.7e+02/ 1.2e+02	1.2e+07 /1.2e+07
sumList tail recursive	F	2.3e+08 /2.3e+08	9.0e+08 /1.0e+09	5.7e+05/ 5.7e+05	1.7e+02 /2.6e+02	3.3e+03 /5.8e+04	1.5e+02 /2.4e+02	1.3e+07/ 1.2e+07
Total		1.1e+09/1.5e+09	4.5e+09/5.6e+09	1.1e+07/4.9e+06	3.1e+05/5.5e+05	1.3e+05/2.0e+05	1.8e+04/1.0e+04	4.3e+07/3.3e+07

Table 30: Per-pass PAPI counters for **List** (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
Total		–/–	–/–	–/–	–/–	–/–	–/–	–/–

Table 31: Per-pass performance for **MonoTree**, ADT has 3 fields, SoA uses 2 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
add1Tree	M	3/3	0%	0.0984	0.1032	0.0796	0.1206	1.24 $\times$	1.05 $\times$	1.30 $\times$	0.86 $\times$
sumTree	F	3/3	0%	0.0092	0.0405	0.0114	0.0806	0.81 $\times$	4.39 $\times$	3.54 $\times$	0.50 $\times$
sumTree TailRec	F	3/3	0%	0.0086	0.0405	0.0109	0.0813	0.79 $\times$	4.72 $\times$	3.71 $\times$	0.50 $\times$
Total				0.1161	0.1841	0.1019	0.2825	1.14 $\times$	1.59 $\times$	1.81 $\times$	0.65 $\times$
Geomean								0.92 $\times$	2.79 $\times$	2.57 $\times$	0.60 $\times$

Table 32: Per-pass PAPI counters for **MonoTree** (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
add1Tree	M	1.6e+08/ 1.5e+08	5.0e+08 /5.9e+08	5.4e+05/ 4.9e+05	4.2e+04 /4.8e+04	8.6e+03/ 6.5e+03	2.5e+03/ 2.2e+03	4.3e+05 /4.4e+05
sumTree	F	4.7e+07 /5.8e+07	2.2e+08 /2.6e+08	1.2e+04 /3.9e+04	3.4e+02/ 3.3e+02	7.0e+02 /4.1e+03	3.0e+02/3.0e+02	8.5e+05/ 7.0e+05
sumTree TailRec	F	4.3e+07 /5.5e+07	1.9e+08 /2.2e+08	6.6e+03 /2.2e+04	3.9e+02/ 2.8e+02	7.3e+02 /4.1e+03	3.5e+02/ 2.6e+02	8.5e+05/ 7.1e+05
Total		2.5e+08/2.6e+08	9.1e+08/1.1e+09	5.6e+05/5.5e+05	4.3e+04/4.8e+04	1.0e+04/1.5e+04	3.2e+03/2.7e+03	2.1e+06/1.8e+06

Table 33: Per-pass PAPI counters for **MonoTree** (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
add1Tree	M	2.4e+08 /5.0e+08	7.1e+08 /1.2e+09	1.5e+05 /1.2e+06	3.4e+04 /5.1e+04	2.3e+03 /2.0e+04	2.3e+03 /2.8e+03	4.1e+05/ 3.9e+05
sumTree	F	2.0e+08 /4.1e+08	4.3e+08 /6.3e+08	8.4e+04 /9.4e+05	2.7e+02 /3.4e+02	1.2e+03 /1.5e+04	2.3e+02 /3.1e+02	3.6e+05 /3.9e+05
sumTree TailRec	F	2.0e+08 /4.1e+08	4.1e+08 /6.4e+08	6.6e+04 /9.3e+05	3.4e+02/ 2.4e+02	1.2e+03 /1.4e+04	2.9e+02/ 2.2e+02	3.6e+05 /4.0e+05
Total		6.5e+08/1.3e+09	1.6e+09/2.5e+09	3.0e+05/3.1e+06	3.5e+04/5.2e+04	4.7e+03/4.9e+04	2.9e+03/3.3e+03	1.1e+06/1.2e+06

Table 34: Per-pass performance for **ObjectGraph**, ADT has 5 fields, SoA uses 4 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
countLargeItems	F	3/5	40%	0.0141	0.0445	0.0155	0.0465	0.91 $\times$	3.16 $\times$	2.88 $\times$	0.96 $\times$
countMarked	F	3/5	40%	0.0185	0.0460	0.0241	0.0480	0.77 $\times$	2.49 $\times$	1.91 $\times$	0.96 $\times$
countSurvivors	F	4/5	20%	0.0139	0.0443	0.0129	0.0468	1.08 $\times$	3.19 $\times$	3.44 $\times$	0.95 $\times$
deadBytes	F	4/5	20%	0.0136	0.0439	0.0123	0.0472	1.11 $\times$	3.22 $\times$	3.58 $\times$	0.93 $\times$
liveBytes	F	4/5	20%	0.0140	0.0434	0.0128	0.0471	1.09 $\times$	3.09 $\times$	3.39 $\times$	0.92 $\times$
sumObjIds	F	3/5	40%	0.0126	0.0435	0.0105	0.0477	1.20 $\times$	3.44 $\times$	4.14 $\times$	0.91 $\times$
sweepUnmarked	M	5/5	0%	0.0719	0.0879	0.0809	0.1867	0.89 $\times$	1.22 $\times$	1.09 $\times$	0.47 $\times$
totalHeapSize	F	3/5	40%	0.0258	0.0713	0.0392	0.0507	0.66 $\times$	2.77 $\times$	1.82 $\times$	1.41 $\times$
touchHotObjects	M	5/5	0%	0.0708	0.0868	0.0827	0.1935	0.86 $\times$	1.23 $\times$	1.05 $\times$	0.45 $\times$
Total				0.2553	0.5116	0.2910	0.7142	0.88 $\times$	2.00 $\times$	1.76 $\times$	0.72 $\times$
Geomean								0.94 $\times$	2.48 $\times$	2.32 $\times$	0.84 $\times$

Table 35: Per-pass PAPI counters for **ObjectGraph** (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
countLargeItems	F	6.4e+07/ 5.2e+07	2.6e+08 /2.8e+08	1.1e+06/ 5.0e+04	2.6e+02 /3.7e+02	3.2e+03 /3.6e+03	2.3e+02 /3.3e+02	2.8e+06/ 6.5e+05
countMarked	F	5.4e+07/ 5.2e+07	2.6e+08 /2.8e+08	1.2e+06/ 5.1e+04	3.7e+02/ 3.7e+02	3.3e+03 /4.0e+03	3.4e+02/ 3.3e+02	2.4e+06/ 5.9e+05
countSurvivors	F	7.0e+07/ 6.5e+07	3.1e+08 /3.8e+08	9.1e+05/ 6.2e+04	2.6e+02 /4.8e+02	2.3e+03 /5.5e+03	2.3e+02 /4.6e+02	2.8e+06/ 1.3e+06
deadBytes	F	6.9e+07/ 6.2e+07	2.7e+08 /3.5e+08	7.3e+05/ 5.4e+04	4.6e+02/ 3.1e+02	2.5e+03 /5.5e+03	4.3e+02/ 2.8e+02	2.8e+06/ 1.3e+06
liveBytes	F	7.1e+07/ 6.2e+07	2.7e+08 /3.5e+08	6.3e+05/ 5.3e+04	3.5e+02 /3.8e+02	2.1e+03 /5.9e+03	3.2e+02 /3.5e+02	2.7e+06/ 1.3e+06
sumObjIds	F	6.4e+07/ 5.3e+07	2.5e+08 /2.7e+08	1.2e+06/ 7.7e+04	2.1e+02 /3.0e+02	3.2e+03 /5.0e+03	1.9e+02 /2.8e+02	2.8e+06/ 6.9e+05
sweepUnmarked	M	1.9e+08 /2.3e+08	5.5e+08 /8.8e+08	1.1e+06/ 3.8e+05	7.2e+04 /1.7e+05	6.1e+03 /1.1e+04	5.6e+03 /5.6e+03	1.9e+06/ 1.0e+06
totalHeapSize	F	4.8e+07 /5.2e+07	2.5e+08 /2.7e+08	1.7e+06/ 1.0e+06	6.3e+02 /7.6e+02	3.5e+04/ 2.1e+04	5.4e+02 /6.0e+02	2.1e+06/ 4.0e+05
touchHotObjects	M	1.8e+08 /2.4e+08	5.7e+08 /9.0e+08	1.2e+06/ 5.0e+05	1.0e+05 /1.4e+05	4.6e+03 /1.1e+04	5.4e+03/ 5.1e+03	1.8e+06/ 9.8e+05
Total		8.1e+08/8.7e+08	3.0e+09/4.0e+09	9.7e+06/2.3e+06	1.8e+05/3.1e+05	6.2e+04/7.2e+04	1.3e+04/1.3e+04	2.2e+07/8.1e+06

Table 36: Per-pass PAPI counters for **ObjectGraph** (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
countLargeItems	F	2.3e+08 /2.4e+08	4.6e+08 /7.9e+08	5.6e+05/ 4.5e+05	3.6e+02/ 2.0e+02	4.9e+03 /8.2e+03	3.3e+02/ 1.8e+02	1.8e+06/ 4.2e+05
countMarked	F	2.3e+08 /2.4e+08	4.6e+08 /7.9e+08	1.3e+06 /1.3e+06	1.3e+02 /4.8e+02	1.8e+04 /2.5e+04	1.2e+02 /4.6e+02	1.8e+06/ 3.6e+05
countSurvivors	F	2.2e+08 /2.4e+08	5.5e+08 /8.2e+08	1.9e+05 /3.6e+05	1.9e+02/ 1.4e+02	1.9e+03 /7.1e+03	1.7e+02/ 1.2e+02	1.8e+06/ 7.1e+05
deadBytes	F	2.2e+08 /2.4e+08	4.9e+08 /8.1e+08	3.8e+05/ 3.7e+05	2.4e+02 /4.2e+02	3.6e+03 /6.9e+03	2.3e+02 /3.8e+02	1.8e+06/ 7.2e+05
liveBytes	F	2.2e+08 /2.4e+08	5.0e+08 /8.1e+08	1.5e+05 /3.6e+05	2.8e+02/ 1.9e+02	1.8e+03 /6.8e+03	2.6e+02/ 1.8e+02	1.8e+06/ 7.3e+05
sumObjIds	F	2.2e+08 /2.4e+08	4.4e+08 /7.8e+08	5.5e+05/ 5.1e+05	3.0e+02 /5.1e+02	4.6e+03 /1.0e+04	2.7e+02 /4.7e+02	1.8e+06/ 4.1e+05
sweepUnmarked	M	2.7e+08 /7.7e+08	7.6e+08 /2.1e+09	1.9e+05 /2.4e+06	7.4e+04 /1.4e+05	3.5e+03 /1.7e+04	3.6e+03 /6.3e+03	1.8e+06/ 9.4e+05
totalHeapSize	F	2.5e+08/ 2.4e+08	4.4e+08 /7.8e+08	3.4e+06/ 1.4e+06	5.2e+02 /6.4e+02	5.9e+04/ 2.6e+04	3.5e+02 /4.8e+02	1.4e+06/ 3.5e+05
touchHotObjects	M	2.7e+08 /8.0e+08	7.7e+08 /2.2e+09	2.4e+05 /1.8e+06	9.7e+04 /1.6e+05	2.8e+03 /2.1e+04	3.3e+03 /4.5e+03	1.7e+06/ 9.4e+05
Total		2.1e+09/3.3e+09	4.9e+09/9.9e+09	6.9e+06/8.8e+06	1.7e+05/3.0e+05	1.0e+05/1.3e+05	8.7e+03/1.3e+04	1.6e+07/5.6e+06

Table 37: Per-pass performance for **PiecewiseFunctions**, ADT has 8 fields, SoA uses 7 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
addConstPW	M	8/8	0%	0.1202	0.1313	0.1374	0.4194	0.88 $\times$	1.09 $\times$	0.96 $\times$	0.31 $\times$
autorefineMaxLevel	F	4/8	50%	0.0202	0.0453	0.0212	0.2345	0.95 $\times$	2.25 $\times$	2.13 $\times$	0.19 $\times$
compressMass	F	3/8	62%	0.0191	0.0459	0.0114	0.2338	1.68 $\times$	2.41 $\times$	4.03 $\times$	0.20 $\times$
diffPW	M	8/8	0%	0.1191	0.1314	0.1317	0.4322	0.90 $\times$	1.10 $\times$	1.00 $\times$	0.30 $\times$
lbDeuxLoadProxy	F	5/8	38%	0.0211	0.0476	0.0186	0.2364	1.14 $\times$	2.26 $\times$	2.57 $\times$	0.20 $\times$
norm2Estimate	F	5/8	38%	0.0320	0.0707	0.0235	0.1981	1.36 $\times$	2.21 $\times$	3.01 $\times$	0.36 $\times$
pmapCutHistogram	F	4/8	50%	0.0317	0.0563	0.0124	0.2420	2.56 $\times$	1.78 $\times$	4.55 $\times$	0.23 $\times$
truncateTolViolations	F	3/8	62%	0.0195	0.0467	0.0114	0.2027	1.70 $\times$	2.40 $\times$	4.09 $\times$	0.23 $\times$
Total				0.3828	0.5753	0.3675	2.199	1.04 $\times$	1.50 $\times$	1.57 $\times$	0.26 $\times$
Geomean								1.31 $\times$	1.85 $\times$	2.42 $\times$	0.25 $\times$

Table 38: Per-pass PAPI counters for `PiecewiseFunctions` (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
addConstPW	M	2.7e+08 /3.5e+08	5.7e+08 /1.2e+09	1.3e+06 /2.0e+06	1.6e+05 /1.9e+05	1.3e+04 /2.2e+04	1.1e+04 /1.8e+04	4.6e+06/ 1.7e+06
autorefineMaxLevel	F	1.0e+08 /1.1e+08	2.5e+08 /3.7e+08	2.5e+06/ 2.8e+05	7.0e+02/ 1.9e+02	1.4e+04/ 6.7e+03	7.1e+02/ 1.7e+02	6.1e+06/ 8.8e+05
compressMass	F	9.7e+07/ 5.8e+07	2.5e+08 /2.8e+08	2.2e+06/ 3.2e+04	8.0e+02/ 4.2e+02	2.0e+04/ 5.2e+03	8.0e+02/ 3.9e+02	6.3e+06/ 7.0e+05
diffPW	M	2.8e+08 /3.4e+08	5.9e+08 /1.3e+09	6.5e+05 /2.4e+06	1.4e+05 /1.7e+05	8.3e+03 /2.3e+04	6.4e+03 /1.8e+04	4.2e+06/ 1.8e+06
lbDeuxLoadProxy	F	1.1e+08/ 9.4e+07	3.3e+08 /4.9e+08	2.4e+06/ 4.4e+05	5.4e+02/ 2.4e+02	2.0e+04/ 8.7e+03	5.4e+02/ 2.2e+02	6.3e+06/ 1.5e+06
norm2Estimate	F	1.2e+08/ 9.3e+07	3.4e+08 /4.4e+08	4.1e+06/ 9.8e+05	6.2e+02/ 4.0e+02	2.0e+05/ 1.8e+04	5.1e+02/ 3.1e+02	5.4e+06/ 1.1e+06
pmapCutHistogram	F	1.6e+08/ 6.3e+07	2.8e+08 /3.7e+08	1.5e+06/ 5.6e+04	9.6e+02/ 4.7e+02	9.3e+04/ 7.5e+03	9.6e+02/ 4.3e+02	4.8e+06/ 1.3e+06
truncateTolViolations	F	9.9e+07/ 5.8e+07	2.5e+08 /2.9e+08	2.1e+06/ 4.7e+04	8.8e+02/ 3.8e+02	1.6e+04/ 4.9e+03	8.4e+02/ 3.6e+02	6.3e+06/ 7.1e+05
Total		1.2e+09/1.2e+09	2.8e+09/4.7e+09	1.7e+07/6.2e+06	3.0e+05/3.6e+05	3.8e+05/9.6e+04	2.2e+04/3.8e+04	4.4e+07/9.7e+06

Table 39: Per-pass PAPI counters for `PiecewiseFunctions` (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
addConstPW	M	3.3e+08 /1.8e+09	7.8e+08 /3.3e+09	2.7e+05 /5.1e+06	1.4e+05 /2.4e+05	8.6e+03 /4.3e+04	5.0e+03 /2.7e+04	4.2e+06/ 1.9e+06
autorefineMaxLevel	F	2.3e+08 /1.2e+09	4.9e+08 /1.3e+09	3.1e+05 /2.1e+06	1.8e+02 /1.4e+03	1.8e+03 /4.0e+04	1.6e+02 /5.7e+02	4.2e+06/ 7.0e+05
compressMass	F	2.3e+08 /1.2e+09	4.7e+08 /1.2e+09	3.0e+05 /1.3e+06	2.6e+02 /1.1e+03	1.9e+03 /3.0e+04	2.4e+02 /2.9e+02	4.1e+06/ 4.0e+05
diffPW	M	3.4e+08 /1.9e+09	8.1e+08 /3.4e+09	2.7e+05 /5.5e+06	1.3e+05 /3.0e+05	5.5e+03 /5.6e+04	5.0e+03 /2.2e+04	4.1e+06/ 1.9e+06
lbDeuxLoadProxy	F	2.4e+08 /1.2e+09	5.3e+08 /1.3e+09	1.6e+05 /2.5e+06	3.6e+02 /2.0e+03	1.8e+03 /4.0e+04	3.3e+02 /1.4e+03	4.2e+06/ 1.0e+06
norm2Estimate	F	3.3e+08 /1.0e+09	5.1e+08 /1.3e+09	4.0e+06/ 2.5e+06	5.8e+02 /1.7e+03	8.0e+04/ 4.1e+04	4.8e+02 /1.3e+03	3.7e+06/ 9.8e+05
pmapCutHistogram	F	2.9e+08 /1.2e+09	5.0e+08 /1.3e+09	7.6e+05 /2.0e+06	9.1e+02/ 7.0e+02	1.0e+04 /3.7e+04	8.8e+02/ 4.5e+02	4.0e+06/ 7.1e+05
truncateTolViolations	F	2.4e+08 /1.0e+09	4.4e+08 /1.3e+09	3.1e+05 /1.3e+06	5.4e+02 /1.4e+03	4.2e+03 /2.7e+04	5.1e+02/ 4.3e+02	4.1e+06/ 4.1e+05
Total		2.2e+09/1.1e+10	4.5e+09/1.4e+10	6.4e+06/2.2e+07	2.7e+05/5.5e+05	1.1e+05/3.2e+05	1.3e+04/5.4e+04	3.3e+07/8.0e+06

Table 40: Per-pass performance for `TernaryTree`, ADT has 5 fields, SoA uses 3 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across -iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
add 1 tree	M	5/5	0%	0.1015	0.1144	0.1135	0.3461	0.89 $\times$	1.13$\times$	1.01 $\times$	0.33 $\times$
sum tree	F	5/5	0%	0.0287	0.0504	0.0280	0.2752	1.03 $\times$	1.76$\times$	1.80$\times$	0.18 $\times$
Total				0.1302	0.1648	0.1414	0.6213	0.92 $\times$	1.27$\times$	1.17$\times$	0.27 $\times$
Geomean								0.96 $\times$	1.41$\times$	1.35$\times$	0.25 $\times$

Table 41: Per-pass PAPI counters for `TernaryTree` (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
add 1 tree	M	2.7e+08/ 2.3e+08	7.5e+08 /9.8e+08	5.8e+05 /1.1e+06	6.8e+04 /1.2e+05	7.7e+03 /1.2e+04	2.5e+03 /2.6e+03	1.5e+06/ 1.1e+06
sum tree	F	1.5e+08/ 1.4e+08	4.6e+08 /5.4e+08	4.4e+04 /2.7e+05	1.2e+02 /1.3e+02	6.8e+02 /8.9e+03	8.6e+01 /1.1e+02	1.8e+06/ 1.2e+06
Total		4.2e+08/3.7e+08	1.2e+09/1.5e+09	6.3e+05/1.3e+06	6.8e+04/1.2e+05	8.4e+03/2.1e+04	2.6e+03/2.7e+03	3.2e+06/2.3e+06

Table 42: Per-pass PAPI counters for `TernaryTree` (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
add 1 tree	M	3.2e+08 /1.6e+09	9.5e+08 /2.4e+09	3.4e+05 /2.3e+06	6.7e+04 /1.8e+05	6.2e+03 /4.4e+04	3.0e+03 /5.1e+03	1.4e+06/ 9.7e+05
sum tree	F	2.6e+08 /1.4e+09	5.8e+08 /1.0e+09	8.0e+04 /2.4e+06	2.5e+02 /1.6e+03	1.4e+03 /4.2e+04	2.3e+02 /6.9e+02	1.4e+06/ 9.5e+05
Total		5.8e+08/3.0e+09	1.5e+09/3.4e+09	4.2e+05/4.6e+06	6.7e+04/1.8e+05	7.6e+03/8.5e+04	3.2e+03/5.8e+03	2.8e+06/1.9e+06

Table 43: Per-pass performance for `Trie`, ADT has 10 fields, SoA uses 9 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across -iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
autocompleteTopKProxy	F	5/10	50%	0.0144	0.0271	0.0422	0.1061	0.34 $\times$	1.88$\times$	0.64 $\times$	0.26 $\times$
countLazyNodes	F	4/10	60%	0.0128	0.0275	0.0401	0.1084	0.32 $\times$	2.14$\times$	0.69 $\times$	0.25 $\times$
countTerminals	F	3/10	70%	0.0125	0.0254	0.0098	0.1007	1.27$\times$	2.03$\times$	2.58$\times$	0.25 $\times$
decayTrieStats	M	10/10	0%	0.0751	0.0803	0.1513	0.3619	0.50 $\times$	1.07 $\times$	0.53 $\times$	0.22 $\times$
resetTraversalState	M	10/10	0%	0.0748	0.0817	0.0836	0.3620	0.89 $\times$	1.09 $\times$	0.98 $\times$	0.23 $\times$
sumPrefixFreq	F	3/10	70%	0.0138	0.0399	0.0117	0.1060	1.18$\times$	2.90$\times$	3.41$\times$	0.38 $\times$
sumSubtreeHints	F	3/10	70%	0.0123	0.0258	0.0098	0.1036	1.25$\times$	2.10$\times$	2.62$\times$	0.25 $\times$
Total				0.2157	0.3076	0.3485	1.249	0.62 $\times$	1.43$\times$	0.88 $\times$	0.25 $\times$
Geomean								0.71 $\times$	1.79$\times$	1.27$\times$	0.26 $\times$

Table 44: Per-pass PAPI counters for `Trie` (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
autocompleteTopKProxy	F	7.3e+07/ 3.9e+07	1.3e+08 /2.1e+08	1.5e+06/ 8.5e+05	1.5e+02 /1.5e+02	4.4e+04/ 1.2e+04	1.4e+02/ 1.3e+02	3.8e+06/ 4.6e+05
countLazyNodes	F	6.5e+07/ 3.1e+07	1.3e+08 /1.6e+08	1.3e+06/ 2.4e+04	1.9e+02 /3.2e+02	1.7e+04/ 3.4e+03	1.8e+02 /3.0e+02	4.1e+06/ 3.6e+05
countTerminals	F	6.1e+07/ 3.0e+07	1.0e+08 /1.2e+08	1.3e+06/ 6.4e+05	2.7e+02/ 2.6e+02	3.0e+04/ 1.3e+04	2.5e+02/ 2.4e+02	4.0e+06/ 2.6e+05
decayTrieStats	M	1.6e+08 /2.1e+08	3.5e+08 /7.9e+08	7.0e+05 /1.0e+06	9.0e+04 /1.2e+05	5.9e+03 /1.8e+04	3.5e+03 /1.6e+04	2.9e+06/ 1.2e+06
resetTraversalState	M	1.6e+08 /2.0e+08	2.8e+08 /7.0e+08	7.8e+05/ 6.1e+05	8.8e+04 /1.5e+05	8.1e+03 /1.2e+04	3.1e+03 /1.3e+04	3.3e+06/ 8.9e+05
sumPrefixFreq	F	5.4e+07/ 3.0e+07	1.1e+08 /1.2e+08	1.7e+06/ 6.4e+05	4.9e+02/ 4.0e+02	2.9e+04/ 1.3e+04	3.8e+02/ 2.6e+02	3.9e+06/ 2.4e+05
sumSubtreeHints	F	6.2e+07/ 3.1e+07	1.1e+08 /1.2e+08	1.4e+06/ 6.4e+05	2.6e+02 /4.5e+02	3.0e+04/ 1.3e+04	2.4e+02 /4.2e+02	4.0e+06/ 2.7e+05
Total		6.4e+08/5.8e+08	1.2e+09/2.2e+09	8.7e+06/4.4e+06	1.8e+05/2.7e+05	1.6e+05/8.5e+04	7.8e+03/3.0e+04	2.6e+07/3.6e+06

Table 45: Per-pass PAPI counters for `Trie` (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
autocompleteTopKProxy	F	1.4e+08 /5.4e+08	2.3e+08 /7.5e+08	1.1e+06 /1.5e+06	1.6e+02 /1.1e+03	3.5e+03 /2.6e+04	1.4e+02 /7.9e+02	2.8e+06/ 4.8e+05
countLazyNodes	F	1.4e+08 /5.5e+08	2.2e+08 /7.6e+08	2.2e+06/ 1.1e+06	2.1e+02 /1.2e+03	2.9e+03 /2.2e+04	2.0e+02 /3.9e+02	2.7e+06/ 3.5e+05
countTerminals	F	1.3e+08 /5.1e+08	2.0e+08 /7.3e+08	1.2e+06/ 6.4e+05	1.7e+02 /6.1e+02	1.9e+03 /1.6e+04	1.6e+02 /2.5e+02	2.9e+06/ 2.0e+05
decayTrieStats	M	1.9e+08 /9.8e+08	4.5e+08 /2.1e+09	2.6e+05 /4.7e+06	8.8e+04 /1.8e+05	4.1e+03 /3.0e+04	3.8e+03 /4.0e+04	2.9e+06/ 1.2e+06
resetTraversalState	M	2.0e+08 /9.7e+08	3.7e+08 /2.0e+09	3.2e+05 /3.3e+06	9.0e+04 /2.3e+05	5.4e+03 /2.9e+04	2.9e+03 /2.8e+04	3.0e+06/ 9.8e+05
sumPrefixFreq	F	1.2e+08 /5.3e+08	2.1e+08 /7.3e+08	1.8e+06/ 6.3e+05	3.6e+02 /9.4e+02	1.1e+04 /1.6e+04	2.3e+02 /4.1e+02	2.5e+06/ 1.9e+05
sumSubtreeHints	F	1.3e+08 /5.3e+08	2.1e+08 /7.3e+08	1.4e+06/ 6.1e+05	2.1e+02 /9.0e+02	1.4e+03 /1.6e+04	2.0e+02 /2.4e+02	2.9e+06/ 2.0e+05
Total		1.0e+09/4.6e+09	1.9e+09/7.8e+09	8.1e+06/1.2e+07	1.8e+05/4.2e+05	3.0e+04/1.6e+05	7.7e+03/7.1e+04	2.0e+07/3.6e+06

Table 46: Per-pass GHC comparison for `OctTree`. Times are median per iteration (s); $\pm$ shows standard error. GHC/Am and GHC/Sm are speedups.

Pass	T	GHC	GHC/Am	GHC/Sm
barnesHutPotential	F	5.346	124.15 $\times$	120.76 $\times$
clearFlags	M	5.304	34.22 $\times$	28.75 $\times$
countActive	F	5.245	118.22 $\times$	168.83 $\times$
countParticles	F	5.304	98.40 $\times$	95.14 $\times$
fmmPotential	F	5.742	44.88 $\times$	26.95 $\times$
scaleEnergy	M	6.062	39.51 $\times$	34.32 $\times$
sumEnergy	F	6.008	148.55 $\times$	159.88 $\times$
sumMass	F	3.229	79.95 $\times$	108.37 $\times$
paletteEntriesQuantized	F	7.070	158.72 $\times$	193.18 $\times$
quantizationErrorProxy	F	4.91e-04	0.01 $\times$	0.01 $\times$
Total		49.31	65.78 $\times$	57.86 $\times$
Geomean		–	33.42 $\times$	33.84 $\times$

Table 47: Per-pass GHC comparison for `Compiler`. Times are median per iteration (s); $\pm$ shows standard error. GHC/Am and GHC/Sm are speedups.

Pass	T	GHC	GHC/Am	GHC/Sm
blockCountPass	F	0.0590	4.15 $\times$	28.57 $\times$
branchStatsPass	F	0.0456	3.27 $\times$	11.66 $\times$
castInstCountPass	F	0.0514	2.55 $\times$	36.46 $\times$
has cycle	F	0.0492	1.32 $\times$	1.46 $\times$
instCountPass	F	0.7260	50.53 $\times$	350.06 $\times$
latencyModelPass	F	0.1253	8.81 $\times$	38.00 $\times$
memoryOpStatsPass	F	0.0561	3.99 $\times$	13.59 $\times$
stripSideEffectsPass	M	0.3612	4.29 $\times$	3.88 $\times$
targetReturnPass	M	0.2512	2.97 $\times$	2.62 $\times$
throughputModelPass	F	0.1007	7.08 $\times$	29.72 $\times$
verifyIR	F	0.3448	19.29 $\times$	84.78 $\times$
Total		2.170	6.60 $\times$	8.78 $\times$
Geomean		–	5.51 $\times$	18.40 $\times$

Table 48: Per-pass GHC comparison for `DBQuery`. Times are median per iteration (s); $\pm$ shows standard error. GHC/Am and GHC/Sm are speedups.

Pass	T	GHC	GHC/Am	GHC/Sm
clearQueryFlags	M	0.5740	7.88 $\times$	6.92 $\times$
countJoins	F	0.0925	3.34 $\times$	8.24 $\times$
filterSelectivitySkew	F	0.1295	4.61 $\times$	10.29 $\times$
hashJoinPressure	F	0.1010	3.56 $\times$	8.08 $\times$
scaleCosts	M	0.7440	8.82 $\times$	6.76 $\times$
sumCost	F	0.5414	20.86 $\times$	25.94 $\times$
sumMemory	F	0.1161	4.22 $\times$	8.17 $\times$
sumRows	F	0.1119	3.24 $\times$	4.96 $\times$
Total		2.410	7.32 $\times$	8.40 $\times$
Geomean		–	5.68 $\times$	8.73 $\times$

Table 49: Per-pass GHC comparison for `DecisionTree`. Times are median per iteration (s); $\pm$ shows standard error. GHC/Am and GHC/Sm are speedups.

Pass	T	GHC	GHC/Am	GHC/Sm
classify Batch	F	8.20e-05	4.56 $\times$	1.32 $\times$
classify Depth	F	1.00e-06	–	–
classify tree	F	4.00e-06	–	2.00 $\times$
countFeatureUses	F	0.1782	2.98 $\times$	2.41 $\times$
countLeaves	F	0.1903	3.32 $\times$	2.86 $\times$
countNodes	F	2.757	46.83 $\times$	40.51 $\times$
countSmallLeaves	F	0.1598	2.66 $\times$	2.23 $\times$
inferenceCost	F	0.1644	2.85 $\times$	2.45 $\times$
sumImpurity	F	0.1595	2.70 $\times$	2.22 $\times$
sumPathLengths	F	0.1797	3.04 $\times$	2.36 $\times$
sumSamples	F	0.1611	2.75 $\times$	2.19 $\times$
treeDepth	F	0.1635	2.80 $\times$	2.43 $\times$
Total		4.113	7.78 $\times$	6.46 $\times$
Geomean		–	3.99 $\times$	2.88 $\times$

Table 50: Per-pass GHC comparison for `DomTree`. Times are median per iteration (s); $\pm$ shows standard error. GHC/Am and GHC/Sm are speedups.

Pass	T	GHC	GHC/Am	GHC/Sm
SumArea	F	0.7979	21.78 $\times$	39.64 $\times$
computeWidths	M	0.1086	3.65 $\times$	2.42 $\times$
count styled	F	0.0516	1.52 $\times$	4.42 $\times$
find max Bottom	F	0.0678	1.98 $\times$	3.68 $\times$
scaleLayout	M	0.0886	3.02 $\times$	2.40 $\times$
sumTextWidth	F	0.0752	2.22 $\times$	6.46 $\times$
Total		1.190	6.02 $\times$	8.29 $\times$
Geomean		–	3.42 $\times$	5.38 $\times$

Table 51: Per-pass GHC comparison for **KDTree**. Times are median per iteration (s); $\pm$ shows standard error. GHC/Am and GHC/Sm are speedups.

Pass	T	GHC	GHC/Am	GHC/Sm
Find nearest Neighbour	F	3.80e-05	0.00 $\times$	0.00 $\times$
countInRange tight_box	F	5.00e-06	0.00 $\times$	0.00 $\times$
photonMappingBenchmark	F	0.9347	20.69$\times$	22.58$\times$
pointCloudNeighborhood	F	0.3075	11.24$\times$	18.01$\times$
sumMassInRange	F	4.00e-06	0.00 $\times$	0.00 $\times$
twoPointCorrelation bin_8_16	F	0.3395	10.78$\times$	16.24$\times$
Total		1.582	7.84$\times$	10.92$\times$
Geomean		–	0.06 $\times$	0.09 $\times$

Table 52: Per-pass GHC comparison for **LinearListReduction**. Times are median per iteration (s); $\pm$ shows standard error. GHC/Am and GHC/Sm are speedups.

Pass	T	GHC	GHC/Am	GHC/Sm
reduction	F	1.264	41.07$\times$	221.89$\times$
Total		1.264	41.07$\times$	221.89$\times$
Geomean		–	41.07$\times$	221.89$\times$